Chapter 15. Multicollinearity and Biased Estimation

Multicollinearity, VIF, ridge regression, principal component regression

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The least squares estimator $\hat\beta = (X'X)^{-1}X'\mathbf{y}$ is the BLUE with covariance $\sigma^2(X'X)^{-1}$. When $X'X$ is nearly singular this covariance is huge, and a slightly biased estimator with much smaller variance can be far more accurate. This chapter studies the problem and two remedies.

Throughout we standardize: y is centred and each regressor is centred and scaled so that $\sum_i x_{ij} = 0$ and $\sum_i x_{ij}^2 = 1$. Then X is $n \times p$ (no intercept column), $X'X$ is the correlation matrix of the regressors, and $X'\mathbf{y}$ is proportional to their correlations with y. Coefficients on the original scale are recovered from $\tilde\beta_j = \hat\beta_j\, s_y/s_j$ and $\tilde\beta_0 = \bar y - \sum_j \tilde\beta_j \bar x_j$, where s_j is the scale used for x_j.

15.1 Multicollinearity

Multicollinearity means that the regressors are (nearly) linearly dependent. Exact dependence makes X rank-deficient and $\hat\beta$ undefined; near dependence makes $(X'X)^{-1}$ huge and $\hat\beta$ unstable.

Example 15.1 (Two regressors)

With two standardized regressors with correlation r_{12},

$$X'X = \begin{pmatrix} 1 & r_{12} \\ r_{12} & 1 \end{pmatrix}, \qquad \operatorname{Var}\hat\beta = \frac{\sigma^2}{1-r_{12}^2}\begin{pmatrix} 1 & -r_{12} \\ -r_{12} & 1 \end{pmatrix},$$

so $\operatorname{Var}\hat\beta_j = \sigma^2/(1-r_{12}^2) \to \infty$ as $|r_{12}| \to 1$, and $\operatorname{Corr}(\hat\beta_1, \hat\beta_2) = -r_{12}$. With $r_{12} = 0.99$ the variances are 50 times those under orthogonal regressors, and confidence intervals $\hat\beta_j \pm t_{\alpha/2}(n-3)\sqrt{MSE/(1-r_{12}^2)}$ are 7 times as wide.

Eigenvalues. If $\lambda_1, \dots, \lambda_p$ are the eigenvalues of $X'X$,

$$\sum_j \operatorname{Var}\hat\beta_j = \sigma^2 \operatorname{tr}(X'X)^{-1} = \sigma^2 \sum_j \frac{1}{\lambda_j},$$

so a single small eigenvalue blows up the total variance. The eigenvector of a small eigenvalue shows which combination of regressors is nearly constant.

Theorem 15.1 (Variance inflation factor)

Let R_j^2 be the coefficient of determination of the regression of x_j on the other regressors. Then c_{jj}, the j-th diagonal entry of $(X'X)^{-1}$, equals $(1-R_j^2)^{-1}$, so

$$\operatorname{Var}\hat\beta_j = \frac{\sigma^2}{1-R_j^2}.$$

Proof. Write $X = (\mathbf{x}_j, X_{-j})$. By the partitioned-inverse formula,

$$c_{jj} = \{\mathbf{x}_j'\mathbf{x}_j - \mathbf{x}_j'X_{-j}(X_{-j}'X_{-j})^{-1}X_{-j}'\mathbf{x}_j\}^{-1} = \{\mathbf{x}_j'(I - H_{-j})\mathbf{x}_j\}^{-1},$$

which is the inverse of the residual sum of squares of $\mathbf{x}_j$ on X_{-j}. Since $\mathbf{x}_j$ is centred with $\mathbf{x}_j'\mathbf{x}_j = 1$, its total sum of squares is 1, so the residual sum of squares is $1 - R_j^2$. ∎

$\mathrm{VIF}_j = (1-R_j^2)^{-1}$ is the **variance inflation factor**: the factor by which $\operatorname{Var}\hat\beta_j$ exceeds its value under orthogonal regressors. $\mathrm{VIF}_j > 10$ (i.e. $R_j^2 > 0.9$) is a common warning sign.

15.2 Ridge regression

Motivation. Under multicollinearity least squares remains unbiased but its variance explodes. Ridge regression adds kI to $X^\top X$; this biases the estimates but, for suitable k, always reduces mean squared error.

Measure the accuracy of an estimator b by the total mean squared error $E\|b-\beta\|^2$. For least squares,

$$E\|\hat\beta-\beta\|^2 = \operatorname{tr}\operatorname{Var}\hat\beta = \sigma^2\sum_j \frac{1}{\lambda_j}.$$

Definition (Ridge estimator)

For $k>0$, $\hat\beta(k)=(X'X+kI_p)^{-1}X'\mathbf{y}$, the minimizer of $\|\mathbf{y}-X\beta\|^2+k\|\beta\|^2$.

Adding k to the diagonal lifts all eigenvalues by k, so $X'X+kI$ is well conditioned. If $X'X=I$, then $\hat\beta(k)=\hat\beta/(1+k)$: ridge **shrinks** least squares towards $\mathbf{0}$. In general $\hat\beta(k)=(X'X+kI)^{-1}X'X\hat\beta$, which is biased: $E\hat\beta(k)=(X'X+kI)^{-1}X'X\beta\neq\beta$.

Canonical form. Let $X'X=P\Lambda P'$ with P orthogonal and $\Lambda=\operatorname{diag}(\lambda_1,\dots,\lambda_p)$. With $Z=XP$ and $\alpha=P'\beta$, the model is $\mathbf{y}=Z\alpha+\epsilon$ with $Z'Z=\Lambda$. Then $\hat\alpha=\Lambda^{-1}Z'\mathbf{y}$, $\operatorname{Var}\hat\alpha=\sigma^2\Lambda^{-1}$, and

$$\hat\alpha(k)=P'\hat\beta(k)=(\Lambda+kI)^{-1}Z'\mathbf{y}, \qquad \hat\alpha_j(k)=\frac{\lambda_j}{\lambda_j+k}\hat\alpha_j :$$

each canonical coefficient is shrunk, most strongly in directions with small λ_j. Since P is orthogonal, $\|\hat\beta(k)-\beta\|=\|\hat\alpha(k)-\alpha\|$.

Theorem 15.2 (Mean squared error of ridge)

$$E\|\hat\beta(k)-\beta\|^2=\sigma^2\sum_{j=1}^{p}\frac{\lambda_j}{(\lambda_j+k)^2}+k^2\sum_{j=1}^{p}\frac{\alpha_j^2}{(\lambda_j+k)^2}.$$

The first term (total variance) decreases in k and the second (squared bias) increases. There is always a $k>0$ with $E\|\hat\beta(k)-\beta\|^2<E\|\hat\beta-\beta\|^2$.

Proof [Supplement]. $\hat\alpha_j(k)$ has mean $\frac{\lambda_j}{\lambda_j+k}\alpha_j$ and variance $\frac{\lambda_j^2}{(\lambda_j+k)^2}\cdot\frac{\sigma^2}{\lambda_j}$; its bias is $-\frac{k}{\lambda_j+k}\alpha_j$. Adding variance and squared bias over j gives the formula. Call it $g(k)$. Then $g(0)=\sigma^2\sum 1/\lambda_j=E\|\hat\beta-\beta\|^2$, and

$$g'(k)=-2\sigma^2\sum_j\frac{\lambda_j}{(\lambda_j+k)^3}+2k\sum_j\frac{\lambda_j\alpha_j^2}{(\lambda_j+k)^3}.$$

At $k=0$, $g'(0)=-2\sigma^2\sum_j\lambda_j^{-2}<0$, so $g(k)<g(0)$ for small $k>0$. ■

In the orthonormal case $X'X=I$, $g(k)=\frac{p\sigma^2+k^2\|\alpha\|^2}{(1+k)^2}$, minimized at $k=p\sigma^2/\|\beta\|^2$ (set $g'(k)=0$). More generally, a separate constant k_j for each canonical direction, $\hat\alpha_j=\frac{\lambda_j}{\lambda_j+k_j}\hat\alpha_j$ (generalized ridge), has mean squared error $\sum_j\frac{\sigma^2\lambda_j+k_j^2\alpha_j^2}{(\lambda_j+k_j)^2}$, minimized at $k_j=\sigma^2/\alpha_j^2$.

Choosing k. The optimal k depends on unknown σ^2 and β. Practical choices:

- the **ridge trace**: plot $\hat\beta_j(k)$ against k and take the smallest k at which the coefficients stabilize;

- the plug-in rule $\hat{k} = p\hat{\sigma}^2/\hat{\beta}'\hat{\beta}$ (Hoerl, Kennard and Baldwin), possibly iterated with $\hat{\beta}(k)$ in place of $\hat{\beta}$;
- cross-validation, e.g. minimizing $\text{PRESS}(k) = \sum_i \{y_i - \mathbf{x}_i'\hat{\beta}_{(i)}(k)\}^2$. As in Theorem 10.3, with $A(k) = X(X'X + kI)^{-1}X' = (a_{ij})$ and $e_i = y_i - \mathbf{x}_i'\hat{\beta}(k)$,

$$\text{PRESS}(k) = \sum_{i=1}^{n} \Big(\frac{e_i}{1 - a_{ii}}\Big)^2,$$

so no refitting is needed (leave-one-out cross-validation).

15.3 Principal component regression

Let $\mathbf{e}_1, \dots, \mathbf{e}_p$ be orthonormal eigenvectors of $X'X$ with eigenvalues $l_1 \geq \cdots \geq l_p > 0$, and $E_r = (\mathbf{e}_1, \dots, \mathbf{e}_r)$. The **principal components** are the columns of $Z_r = XE_r$, and they are orthogonal: $Z_r'Z_r = \text{diag}(l_1, \dots, l_r)$.

All components ($r = p$). $X\beta = XE_pE_p'\beta = Z\alpha$ with $\alpha = E_p'\beta$, and

$$\hat{\beta} = E_p\hat{\alpha} = E_p(Z'Z)^{-1}Z'\mathbf{y} = \sum_{k=1}^{p} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k'X'\mathbf{y}, \qquad \text{Var}\,\hat{\beta} = \sigma^2 \sum_{k=1}^{p} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k',$$

which is just least squares.

Leading components ($r < p$). Regress $\mathbf{y}$ on Z_r only and transform back:

$$\tilde{\beta}_r = E_r(Z_r'Z_r)^{-1}Z_r'\mathbf{y} = \sum_{k=1}^{r} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k'X'\mathbf{y}.$$

Theorem 15.3 (Principal component regression)

$$E\tilde{\beta}_r = \beta - \sum_{k=r+1}^{p} \mathbf{e}_k\mathbf{e}_k'\beta, \qquad \text{Var}\,\tilde{\beta}_r = \sigma^2 \sum_{k=1}^{r} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k',$$

and

$$E\|\tilde{\beta}_r - \beta\|^2 = \sigma^2 \sum_{k=1}^{r} \frac{1}{l_k} + \sum_{k=r+1}^{p} (\mathbf{e}_k'\beta)^2.$$

Proof. Using $X'X = \sum_h l_h\mathbf{e}_h\mathbf{e}_h'$ and orthonormality, $E\tilde{\beta}_r = \sum_{k\leq r} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k'X'X\beta = \sum_{k\leq r} \mathbf{e}_k\mathbf{e}_k'\beta$, and $\beta = \sum_{k\leq p} \mathbf{e}_k\mathbf{e}_k'\beta$. Next,

$$\text{Var}\,\tilde{\beta}_r = \sigma^2 \sum_{k,k'\leq r} \frac{1}{l_k l_{k'}}\mathbf{e}_k\mathbf{e}_k'X'X\mathbf{e}_{k'}\mathbf{e}_{k'}' = \sigma^2 \sum_{k\leq r} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k'.$$

The trace of the variance plus the squared length of the bias gives the mean squared error. ■

Dropping the components with the smallest l_k removes the largest variance terms σ^2/l_k at the cost of the bias $\sum_{k>r}(\mathbf{e}_k'\beta)^2$. This pays off when the small-eigenvalue directions carry little of β. PCR is a "hard" version of ridge: ridge shrinks every canonical coefficient by $l_k/(l_k + k)$, PCR keeps the first r and sets the rest to 0.

Choosing r. Common rules: keep components until the cumulative proportion $\sum_{k\le r} l_k / \sum_k l_k$ (the share of the total variance of the standardized regressors) is large enough; look for an elbow in the scree plot of l_k; or choose r by cross-validation. Note that the components are chosen without looking at y; partial least squares instead builds components that are correlated with y.

15.4 Selected exercises

Motivation. The exercises restate the variance inflation factor on the original scale of the data, compute ridge regression as least squares on augmented data, and show that ridge solves a norm-constrained least squares problem.

Exercise 15.1 (The VIF on the original scale)

In $Y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i$ with uncorrelated errors of variance σ^2 and unstandardized regressors, let $S_{jj} = \sum_i (x_{ij} - \bar{x}_j)^2 > 0$ and let R_j^2 be the coefficient of determination of the regression (with intercept) of x_j on the other regressors. Show that the least squares estimator satisfies

$$\operatorname{Var} \hat{\beta}_j = \frac{\sigma^2}{S_{jj}} \operatorname{VIF}_j, \qquad \operatorname{VIF}_j = (1 - R_j^2)^{-1}.$$

Solution.

Step 1. (Centring.) Let $C = I - J_n/n$, the projection onto $\mathbf{1}^\perp$, and $\tilde{X} = CX_s$, where X_s holds the p regressor columns. By Theorem 6.5 with $X_1 = \mathbf{1}$, the slope vector is $\hat{\beta}_s = (\tilde{X}'\tilde{X})^{-1}\tilde{X}'C\mathbf{y} = (\tilde{X}'\tilde{X})^{-1}\tilde{X}'\mathbf{y}$, so $\operatorname{Var}\hat{\beta}_s = (\tilde{X}'\tilde{X})^{-1}\tilde{X}'(\sigma^2 I)\tilde{X}(\tilde{X}'\tilde{X})^{-1} = \sigma^2(\tilde{X}'\tilde{X})^{-1}$.

Step 2. (Partitioned inverse.) Write $\tilde{X} = (\tilde{\mathbf{x}}_j, \tilde{X}_{-j})$ and $\tilde{H}_{-j} = \tilde{X}_{-j}(\tilde{X}'_{-j}\tilde{X}_{-j})^{-1}\tilde{X}'_{-j}$. As in the proof of Theorem 15.1, the j-th diagonal entry of $(\tilde{X}'\tilde{X})^{-1}$ is $\{\tilde{\mathbf{x}}_j'(I - \tilde{H}_{-j})\tilde{\mathbf{x}}_j\}^{-1}$.

Step 3. (A residual sum of squares.) The column space of $(\mathbf{1}, X_{-j})$ is spanned by $\mathbf{1}$ and the columns of $\tilde{X}_{-j}$, which are orthogonal to $\mathbf{1}$; so its projection is $J_n/n + \tilde{H}_{-j}$, as in the proof of Theorem 10.1(d). Since $\tilde{H}_{-j}\mathbf{1} = \mathbf{0}$, the residual of $\mathbf{x}_j$ on $(\mathbf{1}, X_{-j})$ is

$$(I - J_n/n - \tilde{H}_{-j})\mathbf{x}_j = \tilde{\mathbf{x}}_j - \tilde{H}_{-j}\tilde{\mathbf{x}}_j = (I - \tilde{H}_{-j})\tilde{\mathbf{x}}_j,$$

whose squared length is $\tilde{\mathbf{x}}_j'(I - \tilde{H}_{-j})\tilde{\mathbf{x}}_j$. The total sum of squares of x_j is S_{jj}, so this residual sum of squares equals $S_{jj}(1 - R_j^2)$.

Step 4. Combining, $\operatorname{Var}\hat{\beta}_j = \sigma^2/\{S_{jj}(1 - R_j^2)\}$. The factor σ^2/S_{jj} is the variance in the simple regression on x_j alone (Theorem 4.1(a)), which is also the value when x_j is uncorrelated with the other regressors, and VIF_j is the inflation due to collinearity. Theorem 15.1 is the case $S_{jj} = 1$. ■

Exercise 15.2 (Ridge regression as augmented least squares)

Let $E(\mathbf{y} \mid X) = X\beta$ and $\operatorname{Var}(\mathbf{y} \mid X) = \sigma^2 I_n$, X of size $n \times p$ with singular value decomposition $X = UDV'$ (U of size $n \times p$ with orthonormal columns, $D = \operatorname{diag}(d_1, \ldots, d_p)$ with $d_j \ge 0$, V orthogonal). For $\lambda > 0$ let

$$X_* = \begin{pmatrix} X \\ \sqrt{\lambda} I_p \end{pmatrix}, \qquad \mathbf{y}_* = \begin{pmatrix} \mathbf{y} \\ \mathbf{0}_p \end{pmatrix}, \qquad \hat{\beta} = (X_*'X_*)^{-1}X_*'\mathbf{y}_*.$$

Express $E(\hat{\beta} \mid X)$ and $\operatorname{Var}(\hat{\beta} \mid X)$ through U, D, V, β and σ.

Solution.

Step 1. $X_*'X_* = X'X + \lambda I$ and $X_*'\mathbf{y}_* = X'\mathbf{y}$, so $\hat\beta = (X'X+\lambda I)^{-1}X'\mathbf{y}$ is the ridge estimator with $k=\lambda$; it is defined even if X is rank deficient, because $X'X+\lambda I$ is positive definite.

Step 2. $X'X = VDU'UDV' = VD^2V'$ and $X'X+\lambda I = V(D^2+\lambda I)V'$, so $(X'X+\lambda I)^{-1} = V(D^2+\lambda I)^{-1}V'$ and

$$\hat\beta = V(D^2+\lambda I)^{-1}V'VDU'\mathbf{y} = V(D^2+\lambda I)^{-1}DU'\mathbf{y}.$$

Step 3. (Mean.) $E(\hat\beta \mid X) = V(D^2+\lambda I)^{-1}DU'UDV'\beta = V\,\mathrm{diag}\Big(\dfrac{d_j^2}{d_j^2+\lambda}\Big)V'\beta.$

Step 4. (Variance.) With $M = V(D^2+\lambda I)^{-1}DU'$, $\mathrm{Var}(\hat\beta \mid X) = \sigma^2 MM'$ and $U'U = I$ give

$$\mathrm{Var}(\hat\beta \mid X) = \sigma^2 V(D^2+\lambda I)^{-1}D^2(D^2+\lambda I)^{-1}V' = \sigma^2 V\,\mathrm{diag}\Big(\frac{d_j^2}{(d_j^2+\lambda)^2}\Big)V'.$$

In the coordinates $\alpha = V'\beta$ these are the canonical formulas of Section 15.2 with $\lambda_j = d_j^2$: each component is shrunk by $d_j^2/(d_j^2+\lambda)$. The p extra rows act as pseudo-observations $0 = \sqrt{\lambda}\beta_j + \text{error}$ that pull β towards $\mathbf{0}$. ■

Exercise 15.3 (Ridge as constrained least squares)

Let X $(n\times p)$ have full column rank, $\hat\beta$ be the least squares estimator, $Q(\beta) = (\beta-\hat\beta)'X'X(\beta-\hat\beta)$, and $c>0$.
(a) Find the minimizer of $Q(\beta)$ subject to $\beta'\beta \le c$.
(b) For fixed $k\ge 0$, find the mean and variance of $\hat\beta(k) = (X'X+kI)^{-1}X'\mathbf{y}$.
(c) Using the singular value decomposition $X = UDV'$, show that $\hat\beta(k)$ shrinks $\hat\beta$ towards $\mathbf{0}$.

Solution.

(a) If $\hat\beta'\hat\beta \le c$, then $\hat\beta$ is feasible with $Q(\hat\beta)=0$, and it is the unique minimizer since $X'X$ is positive definite. Assume now $\hat\beta'\hat\beta > c$.

Step 1. For $k\ge 0$, $Q(\beta)+k\|\beta\|^2$ is strictly convex with gradient $2X'X(\beta-\hat\beta)+2k\beta$, which vanishes at $(X'X+kI)^{-1}X'X\hat\beta = (X'X+kI)^{-1}X'\mathbf{y} = \hat\beta(k)$. So $\hat\beta(k)$ is its unique minimizer.

Step 2. With $X'X = P\Lambda P'$ and $\hat\alpha = P'\hat\beta$, $\|\hat\beta(k)\|^2 = \sum_j \lambda_j^2\hat\alpha_j^2/(\lambda_j+k)^2$. This is continuous and strictly decreasing in k (since $\hat\alpha \ne \mathbf{0}$), equals $\|\hat\beta\|^2 > c$ at $k=0$, and tends to 0. Hence there is a unique $k^*>0$ with $\|\hat\beta(k^*)\|^2 = c$.

Step 3. For any feasible $\beta \ne \hat\beta(k^*)$, by Step 1,

$$Q(\beta) > Q(\hat\beta(k^*)) + k^*(\|\hat\beta(k^*)\|^2 - \|\beta\|^2) = Q(\hat\beta(k^*)) + k^*(c-\|\beta\|^2) \ge Q(\hat\beta(k^*)).$$

So the constrained minimizer is the ridge estimator $\hat\beta(k^*)$, the ridge constant being the Lagrange multiplier of the constraint.

(b) $\hat\beta(k) = M\mathbf{y}$ with $M = (X'X+kI)^{-1}X'$, so

$$E\hat\beta(k) = (X'X+kI)^{-1}X'X\beta, \qquad \mathrm{Var}\,\hat\beta(k) = \sigma^2(X'X+kI)^{-1}X'X(X'X+kI)^{-1}.$$

These formulas are for a fixed k; the k^* of (a) depends on $\mathbf{y}$, so they describe the constrained estimator only conditionally on its multiplier.

(c) As in Exercise 15.2, $\hat\beta(k) = V\,\mathrm{diag}\{d_j^2/(d_j^2+k)\}V'\hat\beta$ with all $d_j>0$ by full rank. In the orthonormal coordinates $V'\beta$, each component of $\hat\beta$ is multiplied by $d_j^2/(d_j^2+k) \in (0,1]$, strictly less than 1 for $k>0$; hence $\|\hat\beta(k)\| < \|\hat\beta\|$ whenever $k>0$ and $\hat\beta \ne \mathbf{0}$, and the smallest singular values are shrunk most. ■